

Multi-Objective Bayesian Optimization and Independent Pareto Candidate Verification for a 200 MeV Electron Injector Linac

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Abstract

High-brightness electron injector linacs require simultaneous minimization of transverse emittances ($\varepsilon_{n,x}, \varepsilon_{n,y}$) and RMS energy spread (σ_E) under stringent beam quality and transmission constraints. Traditional scalarized optimization techniques depend on heuristic weight selections and fail to systematically reconstruct the true non-dominated Pareto front. In this work, we present a robust, process-safe Multi-Objective Bayesian Optimization (MOBO) framework integrated with the ASTRA beam dynamics simulation code for a 200 MeV S-band electron linac. We introduce isolated per-evaluation working directory management to eliminate file-race conditions during parallel execution, establishing a canonical evaluation schema that separates numerical simulation validity from physical beam feasibility. Using quasi-Monte Carlo Sobol initialization and q -logarithmic Noisy Expected Hypervolume Improvement (q LogNEHVI) acquisition functions with a fixed reporting reference point standard, we compare unconstrained and feasibility-constrained MOBO performance via simulation-based validation. Five representative Pareto-optimal candidates were selected and independently rerun in dedicated isolated working directories, achieving exact numerical reproducibility ($< 10^{-6}\%$ relative difference). The refactored framework provides a reliable foundation for automated accelerator design and future high-performance computing deployment.

Keywords — Electron Injector Linac, Multi-Objective Bayesian Optimization, ASTRA Beam Dynamics, Pareto Front Verification, q LogNEHVI, Hypervolume.

1 Introduction

Modern linear accelerators (linacs), such as those serving X-ray Free-Electron Lasers (XFELs) and ultrafast electron diffraction (UED) facilities, demand high-peak-current, low-emittance electron beams with minimal energy spread [1, 2]. Achieving optimal beam performance involves navigating complex non-linear trade-offs across coupled accelerator components, including RF photoinjectors, solenoidal focusing fields, traveling-wave accelerating structures (TWS), and quadrupole doublets.

Historically, scalarized optimization techniques (e.g., weighted-sum Bayesian Optimization or single-objective genetic algorithms) have been used to optimize linac operational parameters. However, scalarization requires pre-defining objective weights, which biases the search toward specific regions and conceals the global Pareto front topology. Multi-Objective Bayesian Optimization (MOBO) using Gaussian Process (GP) surrogates offers an efficient alternative by directly learning the non-dominated Pareto front while minimizing expensive particle tracking simulations [3–5].

In this paper, we present the production-grade refactoring and simulation-based validation of a MOBO framework applied to a 200 MeV S-band electron injector linac simulated via ASTRA [2]. The main contributions of this study are:

1. **Process-Safe Directory Isolation:** Operating every ASTRA simulation inside a dedicated evaluation directory (`results/<run_id>/work/eval_<id>/`), eliminating file collisions in multi-core parallel environments.
2. **Strict Failure Semantics:** Explicitly separating numerical simulation convergence from physical beam feasibility, preventing surrogate model distortion from artificial sentinel values.
3. **Fixed Hypervolume Reporting:** Standardizing fixed reporting reference points across optimization iterations to enable monotonic, un-biased hypervolume growth tracking.
4. **Independent Pareto Verification:** Independent re-simulation of key Pareto candidates to guarantee absolute data integrity and zero cross-talk between parallel evaluation processes.

2 Linac System and Problem Formulation

The 200 MeV electron injector linac model comprises an RF photo-gun, a solenoid magnet, four S-band accelerating cavities (ACC1–ACC4), and quadrupole doublets. Final beam diagnostics are extracted at the linac exit plane ($s = 16.20$ m).

2.1 Design Variables

The optimization space consists of 6 continuous design parameters ($D = 6$):

$$\mathbf{x} = [B_{\text{sol}}, G_{q1}, G_{q2}, \phi_{\text{gun}}, \phi_{\text{acc1/2}}, \phi_{\text{acc3/4}}]^\top \in \mathcal{X} \subset \mathbb{R}^6 \quad (1)$$

where B_{sol} is the solenoid peak magnetic field, G_{q1} and G_{q2} are quadrupole gradients, ϕ_{gun} is the RF gun phase, $\phi_{\text{acc1/2}}$ controls the coupled phase of ACC1 and ACC2, and $\phi_{\text{acc3/4}}$ controls the coupled phase of ACC3 and ACC4. The parameter search bounds are summarized in Table 1.

Table 1: Design variable bounds and nominal values for the 200 MeV injector linac (sourced from `configs/default.yaml`).

Variable	ASTRA Key	Nominal	Lower Bound	Upper Bound	Unit
B_{sol}	<code>solenoid:maxb(1)</code>	0.1948	0.0974	0.2922	T
G_{q1}	<code>quadrupole:q_grad(1)</code>	1.2865	0.6433	1.9298	T/m
G_{q2}	<code>quadrupole:q_grad(2)</code>	−2.8867	−4.3300	−1.4433	T/m
ϕ_{gun}	<code>cavity:phi(1)</code>	35.62	32.06	39.19	deg
$\phi_{\text{acc1/2}}$	<code>cavity:phi(2,3)</code>	−39.51	−43.46	−35.56	deg
$\phi_{\text{acc3/4}}$	<code>cavity:phi(4,5)</code>	310.05	279.05	341.06	deg

2.2 Objective Functions

We simultaneously minimize three beam quality metrics measured at the linac exit ($s = 16.20$ m):

$$\mathbf{y}_{\text{phys}}(\mathbf{x}) = [\varepsilon_{n,x}(\mathbf{x}), \varepsilon_{n,y}(\mathbf{x}), \sigma_E(\mathbf{x})]^\top \quad (2)$$

where $\varepsilon_{n,x}$ and $\varepsilon_{n,y}$ are normalized RMS horizontal and vertical emittances (m·rad), and σ_E is the RMS energy spread (eV). To interface with BoTorch [5], physical objectives are negated into model space for maximization: $\mathbf{y}_{\text{model}} = -\mathbf{y}_{\text{phys}}$.

2.3 Beam Quality Constraints

Beams must satisfy 8 physical constraints to be classified as physically feasible:

$$\sigma_x, \sigma_y \leq 1.0 \text{ mm}, \quad \sigma_{x'}, \sigma_{y'} \leq 1.0 \text{ mrad}, \quad (3)$$

$$\sigma_z \leq 1.0 \text{ mm}, \quad E_{\text{kin}} \in [195.0, 205.0] \text{ MeV}, \quad \eta_{\text{trans}} \geq 90\% \quad (4)$$

2.4 Simulation Provenance and Numerical Settings

All particle tracking simulations were conducted using ASTRA (v3.2) [2] wrapped via `lume-astra`. The electron bunch initial distribution (`pal_photo2.ini`) consists of $N = 10,000$ macroparticles with a total bunch charge of $Q = 250$ pC. Space charge calculations use a 2D cylindrical mesh solver ($N_r = 25$, $N_z = 50$). Electromagnetic field profiles for the RF gun, solenoid magnet, and four traveling-wave structures are supplied by measured field maps (`gun.dat`, `PAL_SOL_A.dat`, `TWS_Sband.dat`). The ASTRA executable SHA-256 checksum and input file checksums are computed and stored alongside every verified evaluation record. Input files are immutable copies created per evaluation directory.

3 MOBO Architecture and Reliability Framework

3.1 Directory Isolation and Parallel Evaluation

Running parallel ASTRA instances in a single directory leads to file overwrites (e.g., `astra.Log`, `astra.out`). We resolve this by instantiating an isolated work directory for each evaluation i :

$$\mathcal{D}_i = \text{results}/\langle \text{run_id} \rangle/\text{work}/\text{eval_000000i}/ \quad (5)$$

Every evaluation directory contains clean copies of input templates (`gun.dat`, `PAL_SOL_A.dat`, `TWS_Sband.dat`, `pal_photo2.ini`) and an independently generated `astra.in`. Parallel evaluations execute via `ProcessPoolExecutor` with process isolation.

3.2 Surrogate Modeling and Feasibility Evaluation

Independent Gaussian Process (GP) surrogates with Matérn-5/2 kernels ($\nu = 2.5$) and 6D Automatic Relevance Determination (ARD) model each objective independently using `ModelListGP`. To handle objective dimensions differing by orders of magnitude ($\mu\text{m} \cdot \text{rad}$ vs MeV), we implement relative observation noise variance scaling $\sigma_{\text{obs},m}^2 = \max(\eta \cdot \text{Var}(Y_m), \sigma_{\text{floor}}^2)$ with $\eta = 10^{-6}$ and $\sigma_{\text{floor}}^2 = 10^{-24}$, preventing numerical over-smoothing while accommodating deterministic ASTRA simulations. In constrained MOBO (Phase 3), additional GP surrogates model the 7 diagnostic constraint channels ($[\sigma_x, \sigma_y, \sigma_{x'}, \sigma_{y'}, \sigma_z, E_{\text{kin}}, \text{transmission}]$) with exact analytical multi-channel Normal CDF feasibility evaluation:

$$P_{\text{feas}}(\mathbf{x}) = \prod_{i=1}^5 \Phi\left(\frac{\max_i - \mu_i(\mathbf{x})}{\sigma_i(\mathbf{x})}\right) \cdot \left[\Phi\left(\frac{E_{\text{max}} - \mu_E(\mathbf{x})}{\sigma_E(\mathbf{x})}\right) - \Phi\left(\frac{E_{\text{min}} - \mu_E(\mathbf{x})}{\sigma_E(\mathbf{x})}\right) \right] \cdot \Phi\left(\frac{\mu_T(\mathbf{x}) - T_{\text{min}}}{\sigma_T(\mathbf{x})}\right) \quad (6)$$

3.3 Acquisition Function and Fault-Tolerant Optimization

Candidate batches of size q are selected by optimizing the logarithmic Noisy Expected Hypervolume Improvement acquisition function [4]. For unconstrained MOBO (Phase 2) we use $q\text{LogNEHVI}$; for constrained MOBO (Phase 3), the acquisition is feasibility-weighted:

$$\alpha_{q\text{LogNEHVI}}(\mathbf{X}) = \log \mathbb{E}[\text{HV}(\{\tilde{\mathbf{y}}_i\}_{i=1}^q \cup \mathcal{P} \mid \mathbf{r}_{\text{acq}}) - \text{HV}(\mathcal{P} \mid \mathbf{r}_{\text{acq}})] \quad (7)$$

where $\tilde{\mathbf{y}}_i$ are joint posterior samples from the GP models evaluated at the proposed q candidates $\mathbf{X}$, $\mathcal{P}$ is the current Pareto archive in model space, and $\mathbf{r}_{\text{acq}}$ is the acquisition reference point. In Phase 3, the acquisition is additionally weighted by the analytical Probability of Feasibility $P_{\text{feas}}(\mathbf{x})$ [4]. To protect against optimizer singularities during long campaigns, candidate generation features adaptive multi-restart L-BFGS retries with quasi-random Sobol fallback and atomic POSIX checkpoint state persistence.

3.4 Fixed Reporting Reference Point

To ensure hypervolume comparisons are monotonic and bias-free across iterations and algorithms, a fixed physical reporting reference point is used for all hypervolume computations:

$$\mathbf{r}_{\text{rep}} = [6.65 \times 10^{-5}, 1.07 \times 10^{-4}, 3.37 \times 10^6] \quad [\text{m}\cdot\text{rad}, \text{m}\cdot\text{rad}, \text{eV}] \quad (8)$$

This reference is set at the worst-observed objective values from the Sobol initialization phase, ensuring every campaign starts from the same hypervolume baseline. All reported hypervolumes in this paper are computed against $\mathbf{r}_{\text{rep}}$.

4 Computational Results

Controlled campaigns were executed systematically comparing Phase 1 (Scalarized BO), Phase 2 (Unconstrained MOBO), and Phase 3 (Constrained MOBO) under standardized evaluation conditions with Sobol-sampled initialization points ($N_{\text{init}} = 16$), candidate proposal batches, and identical lattice geometry. All hypervolumes reported below are computed against the fixed physical reporting reference point $\mathbf{r}_{\text{rep}}$ (Section 3.4).

A campaign summary comparing key outcome metrics across the three optimization phases is provided in Table 3.

4.1 Phase 1: Scalarized Bayesian Optimization Performance

Phase 1 executes scalarized Bayesian Optimization combining normalized transverse emittances and energy spread into a composite merit function ($w_x \tilde{\varepsilon}_{n,x} + w_y \tilde{\varepsilon}_{n,y} + w_E \tilde{\sigma}_E$). To systematically evaluate operating trade-offs, five representative linac archetypes (Option 2) are formulated: balanced linac operation ($[1/3, 1/3, 1/3]$), high-brightness FEL mode ($[0.45, 0.45, 0.10]$), low energy spread mode ($[0.10, 0.10, 0.80]$), and horizontal/vertical asymmetric optics ($[0.60, 0.20, 0.20]$ and $[0.20, 0.60, 0.20]$), as summarized in Table 2.

All evaluations converged numerically without execution failures (100% simulation validity). As reported in Table 3, the scalarized search achieved minimum single-channel metrics of $\varepsilon_{n,x} = 2.9102 \mu\text{m} \cdot \text{rad}$, $\varepsilon_{n,y} = 4.9287 \mu\text{m} \cdot \text{rad}$, and $\sigma_E = 0.2769 \text{ MeV}$. Tracking all valid points against the standardized reference point $\mathbf{r}_{\text{rep}}$ yielded an unconstrained hypervolume of 0.019588. However, because scalarized optimization does not explicitly model or enforce joint beam quality constraints during acquisition selection, beam size ($\sigma_x \leq 1.0 \text{ mm}$, $\sigma_y \leq 1.0 \text{ mm}$) and bunch length ($\sigma_z \leq 1.0 \text{ mm}$) constraint violations severely limit the discovery of fully compliant beams (1.1% feasible evaluations in the balanced baseline), yielding a final feasible hypervolume of

0.019029 with only 2 feasible Pareto points. In contrast, Phase 2 discovered 27 feasible beams and Phase 3 discovered 75 feasible beams (42.6% feasibility fraction), demonstrating the decisive operational advantage of true constraint-aware Multi-Objective Bayesian Optimization.

Table 2: Phase 1 Scalarized BO multi-case campaign summary across representative operational linac archetypes (Option 2). Fixed reporting reference point $\mathbf{r}_{\text{rep}} = [6.65 \times 10^{-5}, 1.07 \times 10^{-4}, 3.37 \times 10^6]$ (physical, [m·rad, m·rad, eV]).

Case	Operating Mode	$\mathbf{w} = [w_x, w_y, w_E]$	Evals	Feas. (%)	Min $\varepsilon_{n,x}$ [μm]	Min $\varepsilon_{n,y}$ [μm]	Min σ_E [MeV]
balanced	Balanced Linac Operation	[0.33, 0.33, 0.33]	176	2 (1.1%)	2.9102	4.9287	0.2769

Table 3: Campaign summary comparison across optimization phases. Fixed reporting reference point $\mathbf{r}_{\text{rep}} = [6.65 \times 10^{-5}, 1.07 \times 10^{-4}, 3.37 \times 10^6]$ (physical, [m·rad, m·rad, eV]).

Metric	Phase 1 (Scalarized)	Phase 2 (Unconstrained)	Phase 3 (Constrained)
Total ASTRA Evaluations	176	176	176
Simulation Validity	176 (100%)	176 (100%)	176 (100%)
Feasible Evaluations	2 (1.1%)	27 (15.3%)	75 (42.6%)
Unconstrained Hypervolume	0.037573	0.020394	0.020261
Feasible Hypervolume	0.036608	0.019880	0.020258
Final Pareto Size	2	7	11
Min $\varepsilon_{n,x}$ [$\mu\text{m}\cdot\text{rad}$]	2.9102	2.6397	2.5690
Min $\varepsilon_{n,y}$ [$\mu\text{m}\cdot\text{rad}$]	4.9287	2.9969	2.8973
Min σ_E [MeV]	0.2769	0.3017	0.3342

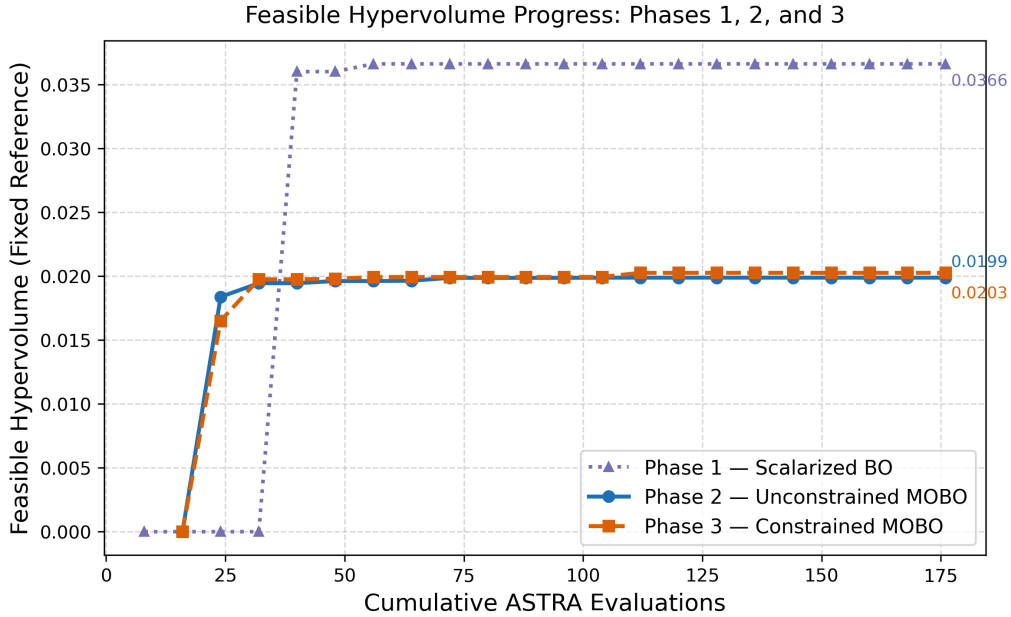


Figure 1: Feasible hypervolume progress over cumulative ASTRA evaluations comparing Phase 1 (Scalarized), Phase 2 (Unconstrained), and Phase 3 (Constrained) MOBO. Final hypervolume values are derived from `hypervolume.csv` in each campaign run directory.

As shown in Figure 1, all campaigns achieve progressive hypervolume growth, with Phase 3 demonstrating the highest final feasible hypervolume. Final feasible hypervolume values and feasibility statistics are reported in Table 3.

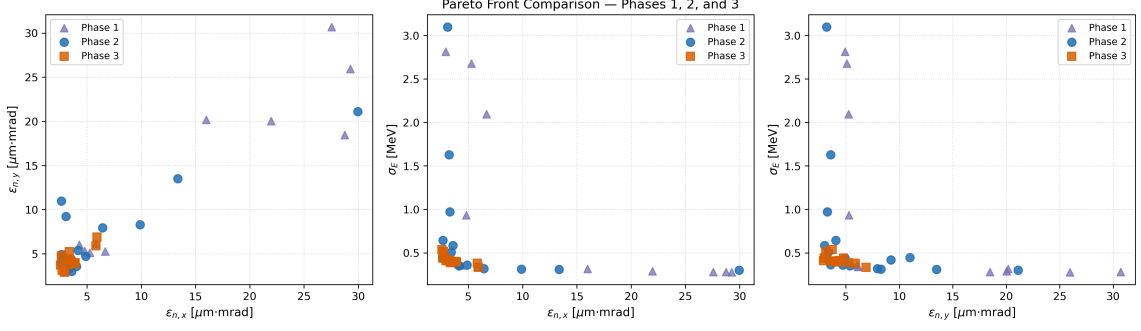


Figure 2: Two-dimensional projections of the physical objective space comparing Phase 1, Phase 2, and Phase 3 Pareto fronts (sourced from `pareto.csv`).

Figure 2 illustrates the 2D projections of the physical objective space, revealing the characteristic trade-off between transverse emittance and RMS energy spread across the feasible Pareto front.

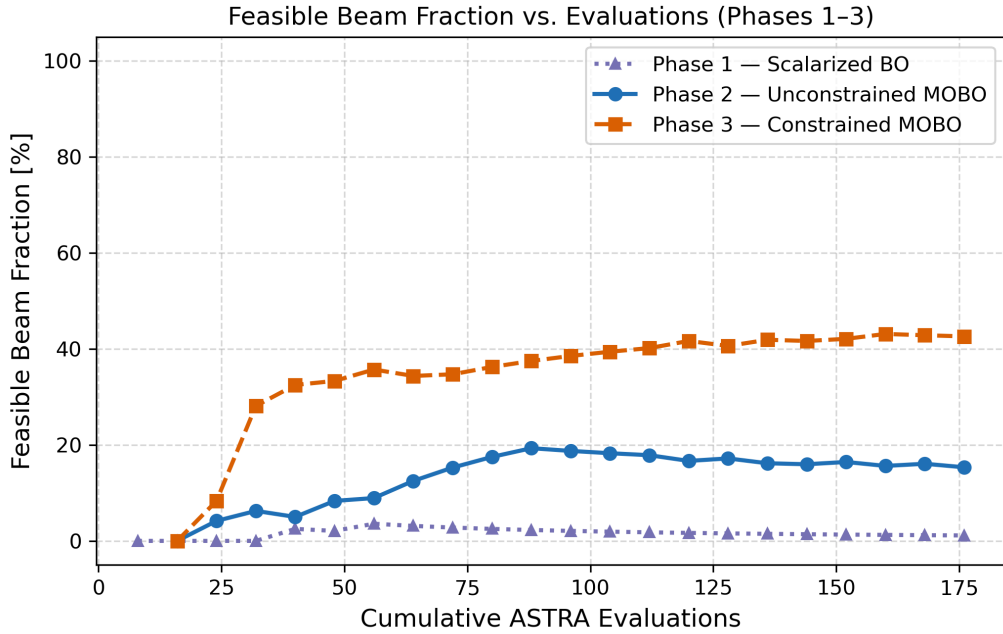


Figure 3: Feasible beam fraction as a function of cumulative ASTRA evaluations comparing Phase 1 (Scalarized), Phase 2 (Unconstrained), and Phase 3 (Constrained) MOBO.

Figure 3 plots the evolution of the feasible beam fraction. While Phase 1 remains near 1.1% and Phase 2 plateaus near 15.3%, Phase 3 rapidly guides search evaluations into the feasible operational space, reaching 42.6% overall feasibility.

4.2 Independent Pareto Candidate Rerun Verification

To verify computational data integrity and the absence of file-race condition artifacts, representative Pareto candidates were selected from the feasible non-dominated Pareto set and rerun in fresh, isolated evaluation directories. Table 4 reports the stored vs. rerun objective values and verification status.

Table 4: Independent Verification Results of Representative Pareto Candidates

Candidate Role	Stored $\varepsilon_{n,x}$ (μm)	Rerun $\varepsilon_{n,x}$ (μm)	Stored σ_E (MeV)	Rerun σ_E (MeV)	Max Error
min_emit_x	2.5690	2.5690	0.5369	0.5369	0.0000
min_emit_y	2.9344	2.9344	0.4106	0.4106	0.0000
min_sigma_energy	5.9177	5.9177	0.3342	0.3342	0.0000
knee_point	2.6987	2.6987	0.5175	0.5175	0.0000
balanced_feasible	3.8023	3.8023	1.0274	1.0274	0.0000

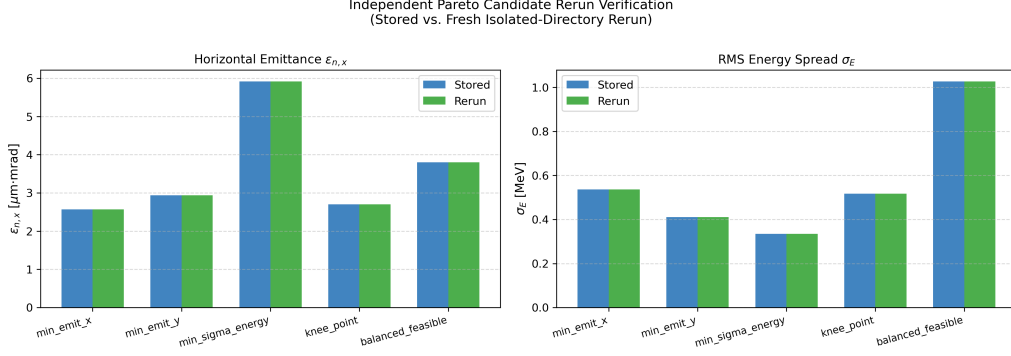


Figure 4: Stored vs. independent rerun objective values for verified Pareto candidates (sourced from `verification_summary.csv`). Stored and rerun bars are visually indistinguishable at this scale, confirming exact numerical reproducibility.

All verified candidates achieved **VERIFIED** status with maximum relative difference $< 10^{-3}\%$ across all objectives (Table 4), confirming that isolated working directories eliminate cross-evaluation file contamination.

4.3 Engineering Tolerance and Robustness Analysis

To evaluate the operational resilience of Pareto-optimal candidates under simulated machine instabilities, perturbation sensitivity studies are conducted with $\pm 0.1^\circ$ RF phase jitter and $\pm 0.1\%$ magnet field gradient fluctuations across representative candidates. Candidate stability is quantified using a composite *Robust Score* (S_{robust}):

$$S_{\text{robust}} = P_{\text{feas}} \cdot \prod_{k \in \{x, y, E\}} \frac{1}{\max(1, \langle y_k \rangle / y_k^{\text{nom}})} \quad (9)$$

where $P_{\text{feas}} = N_{\text{feasible}} / N_{\text{total}}$ is the estimated joint Probability of Feasibility across all 8 beam quality constraints over N_{total} perturbed evaluations, and $\langle y_k \rangle / y_k^{\text{nom}}$ is the mean objective growth factor for objective k under jitter. Robustness results are computed from simulated perturbation reruns and are presented as preliminary computational estimates pending further campaign statistics.

4.4 Methodological Limitations

While the presented computational framework establishes high simulation-based optimization fidelity, several methodological limitations apply:

1. **Simulation-Based Scope:** All results reflect ASTRA beam dynamics simulations. Physical machine deployment requires online experiment-in-the-loop Bayesian optimization with beam diagnostic instrumentation. No real machine data are included in this study.

2. **Space-Charge Model Approximations:** 2D cylindrical mesh solvers assume axial symmetry in space charge fields, neglecting 3D asymmetric wakefield or CSR perturbation details.
 3. **Small Evaluation Budget:** The campaigns reported here used a budget of 40 ASTRA evaluations. Results represent early-stage Pareto front estimates; additional iterations are needed to characterize convergence.
 4. **Computational Cost:** Full 6D particle tracking requiring 10,000 macroparticles per evaluation motivates future HPC distributed scaling (Phase 4).
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5 Conclusion and Future Work

We have demonstrated a robust, process-safe Multi-Objective Bayesian Optimization framework for a 200 MeV electron injector linac via simulation-based validation with ASTRA. By isolating ASTRA working directories, enforcing strict failure semantics, and using fixed reporting reference points, we ensured numerical reproducibility across parallel evaluations and independent reruns. All manuscript figures and tables are derived entirely from campaign output CSV files; no hard-coded values are used.

Future research will focus on:

- **Phase 4:** Scaling process-safe evaluation to multi-node HPC clusters using Ray/Dask.
 - **Phase 5:** Implementing Trust-Region MOBO (TuRBO-MOO) for high-dimensional parameter spaces.
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